

PROMIT MUKHERJEE

AI / ML RESEARCHER & SYSTEMS BUILDER

Bengaluru, Karnataka, India | 6290761565 | promitmukherjeeofficial@gmail.com | [GitHub](#) | [LinkedIn](#)

SUMMARY

MCA student focused on machine learning, computer vision, and representation learning, with additional interest in algorithms and systems-level implementation. Explores technical problems through experimentation and first-principles implementation across medical imaging, computational biology, neural representation learning, and memory-efficient algorithms.

EDUCATION

Master of Computer Applications (MCA) — RNS Institute of Technology, Bengaluru

2025 – 2027

CGPA: 8.60

RESEARCH PROJECTS

Diabetic Retinopathy AI — *Medical AI / Computer Vision / Self-Supervised Learning*

- Developing a JEPA-based self-supervised representation learning approach with a ResNet101 encoder for 5-level diabetic retinopathy grading across 35,126 fundus images.
- Observed QWK improvement from 0.78 to 0.84 and Macro-F1 from 0.71 to 0.76 through iterative model refinement.

Quantum-Inspired RNA Storage — *Computational Biology / Information Theory*

- Developing a quantum-inspired computational approach for encoding data into RNA base sequences under biological sequence constraints, with simulated decoding and error-resilience.

MatrixResearch v1 — *Representation Learning / Neural Matrix Representation*

- Designed an autoencoder (encoder $256 \rightarrow 1024 \rightarrow 512 \rightarrow 256 \rightarrow 128$, decoder $128 \rightarrow 256 \rightarrow 512 \rightarrow 1024 \rightarrow 256$) for neural matrix representation, evaluated across datasets of 1,000, 5,000, and 10,000 matrices.
- Found that increasing dataset diversity exposed latent representation collapse in the MLP bottleneck, with latent standard deviation dropping from ≈ 0.182 to $\approx 0.007-0.028$.

Stable In-Place Merge — *Algorithms / Memory-Efficient Computing*

- Implemented a stable, in-place merge algorithm achieving $O(1)$ auxiliary memory, $O(m \log(n/m + 1))$ comparisons, and $O((m+n)V_m)$ assignments via distinct buffer extraction, block rearrangement, and Hwang–Lin local merge.
- Validated correctness through 100,000+ randomized payload tests, property-based testing, fuzzing, ASAN/UBSAN, and cross-platform CI.

PROJECTS

Smart Attendance System using ML — github.com/PromitMukherjee/Smart-Attendance-System-using-ML

- Built a computer-vision attendance application using OpenCV and machine-learning-based facial processing. (Python, OpenCV, Machine Learning, Computer Vision)

Machine Learning Algorithms — github.com/PromitMukherjee/Machine_learning_Algorithm

- Implemented core machine-learning algorithms from first principles to develop foundational understanding.

TECHNICAL SKILLS

Programming: Python, C++, C++20

Machine Learning: PyTorch, Scikit-learn, OpenCV

Algorithms & Systems: Algorithms, Memory-Efficient Systems

Tools: SQL, Git, Linux, ASAN, UBSAN

ACHIEVEMENTS

VisionX — Internal College Hackathon, RNS Institute of Technology (2025–26)

Runner-up